Kazakhstan regional birth rate: now and next quarter

ECO1209 Data Analysis and Economic Forecasting, KBTU. Group assignment “Build an Indicator”. Team: [names]. Model estimated 2026-10-18T17:30:33+00:00; data retrieved 2026-10-18. Repository: github.com/I2Rchik/KZ-birth-rates.

1. Indicator card

Element	Statement
Concept	How many babies are born per 1,000 residents of each region this quarter, and what next quarter is likely to bring. Regional authorities use it to plan maternity capacity, kindergarten and school places.
Formula	$CBR(i,q) = 4 \times 1{,}000 \times B(i,q) / \bar{P}(i,q)$. B = live births registered in quarter q in region i. $\bar{P}$ = mean of the populations at the start of q and of q+1. The factor 4 annualises the rate.
Unit and scale	Live births per 1,000 population, annualised. Changes are year-on-year, in births per 1,000 and in percent.
Frequency and lag	Quarterly. Published about 40–45 days after the quarter ends (January–March 2025 data were released on 12 May 2025).
Coverage	16 harmonised regions on pre-2018 borders (14 oblasts, Astana, Almaty city). 2014Q1–2026Q2.
Source and vintage	Bureau of National Statistics, Taldau: “Число родившихся (живыми)”, “Численность населения на начало периода”. Retrieved 18 October 2026. Monthly operational figures are revised when the annual data are final.
Transformations	Cumulative January–quarter-end births are differenced to quarters after harmonising regions. Abai, Zhetisu and Ulytau (2022) and Shymkent (2018) are added to their parent regions. Taking logs makes changes read as percentages. The four-quarter difference removes seasonality without a seasonal-adjustment model.
Known weaknesses	A crude rate ignores age structure. Births are counted by registration date. The source has no January–June 2022 figure, so 2022Q2–Q3 are blank. Population from 2024Q1 onward is projected (the downloaded series ends in January 2024). Harmonised units hide the 2022 regions.

2. Data and pipeline

Series	Taldau indicator (as exported)	File in data/raw
births	Число родившихся (живыми), человек; Всего; январь–квартал нарастающим итогом	births_quarterly_cumulative.csv.gz
population	Численность населения на начало периода, человек; Всего; месяц	population_monthly.csv.gz
marriages_annual	Число зарегистрированных браков, человек; Всего; год (КСП 612401)	marriages_annual.csv.gz
income_annual	Среднедушевые номинальные денежные доходы населения, тенге; год	income_per_capita_annual.csv.gz
housing_pc	Обеспеченность жильем на одного проживающего, м2; Всего; год	housing_per_resident_annual.csv.gz
cpi	Индекс потребительских цен, %, товары и услуги (ГНКИПЦ01_2753); отчетный период к предыдущему периоду; квартал	cpi_quarterly_by_region.xls
marriages	Number of registered marriages (Число зарегистрированных браков), persons; Total; January–quarter end, cumulative (English export)	marriages_quarterly_cumulative.csv.gz
income	Average per capita nominal monetary income of the population (Среднедушевые номинальные денежные доходы), tenge; quarterly (English export)	income_per_capita_quarterly.csv.gz

All six files are untouched Taldau exports with SHA-256 checksums in `data/raw/manifest.json`. The build stops if any file changes. The data layer (`src/ingest.py`) keeps only oblasts and the three republican cities. It drops districts, towns, villages, the Alatau city administration and the national total, which Taldau lists in the same files. Check: the 16 regions sum exactly to the national births total in every quarter. Where Taldau reports old South Kazakhstan for a period, that row replaces Turkestan and Shymkent, so Shymkent is not counted twice in 2017–18. Population is summed across regions month by month, then averaged per quarter. Annual drivers (marriages, per-capita income, housing per resident) are dated at the quarters of the following year, when they are published. The CPI is chained from quarter-on-quarter indices. **One command** rebuilds everything: `python run.py rebuild`. Deleting `data/processed` and re-running restores it byte for byte, which a test checks.

3. Model and results

Dependent variable: $y = \ln CBR(t) - \ln CBR(t-4)$. Estimator: panel ARDL with region fixed effects, by OLS, with Driscoll-Kraay, Bartlett kernel, 3 lags standard errors. These are robust to the common shocks shared by all regions (Driscoll & Kraay, 1998; Hoechle, 2007). Sample 2016Q1–2026Q2, 16 regions, 30 quarters, 480 region-quarters. Within $R^2 = 0.39$. Expected signs were fixed before estimation. The last column gives the effect of a one-standard-deviation move, in births per 1,000 at the sample mean.

Regressor	Exp.	Coef.	DK s.e.	p	1 s.d. effect
Own momentum (lag 1)		0.639	0.086	<0.001	0.94

Regressor	Exp.	Coef.	DK s.e.	p	1 s.d. effect
Own momentum (lag 4)		-0.102	0.077	0.195	-0.14
Marriage rate (yoy log change), lag 3	+	-0.041	0.015	0.012	-0.16
Real per-capita income (yoy log change), lag 4	+	0.116	0.068	0.098	0.16
Housing per resident, last published year (yoy log change), lag 1	+	-0.114	0.246	0.648	-0.04

Diagnostics. Serial correlation (lags 1–4): p <0.001. Breusch–Pagan: p 0.004. Jarque–Bera: p <0.001. Max VIF 1.31. Pesaran CD: p <0.001, mean cross-region residual correlation 0.47. Chow stability: 2018Q2: p <0.001; 2020Q2: p <0.001; 2022Q3: p <0.001, so stability is rejected at every break date.

Integration. For ln CBR, ADF rejects in 0% of regions and KPSS rejects in 88%: unit root (both tests agree). For y, ADF rejects in 0% and KPSS in 0% (Fisher-ADF p 0.406): inconclusive (tests disagree). No regression is run in levels, so a spurious levels regression cannot arise. Engle–Granger (ln CBR on ln marriage rate) is reported in Annex A for information only.

Out of sample. Estimated once on 2016Q1–2024Q4, then one-step-ahead forecasts for 2025Q1–2026Q2 (96 region-quarters). RMSE: model 1.672 vs benchmark 2.059 (ratio 0.812). MAE: 1.383 vs 1.797 (ratio 0.770). The benchmark is a seasonal random walk, chosen in advance. **The model wins**, but the Diebold–Mariano test gives p 0.073, so the gain is not significant at 5%. Falsification rule 1 (beat the benchmark): passed. Rule 2 (marriage effect > 0 at 5%, one-sided): FAILED (one-sided p 0.994).

Economic size. A one-standard-deviation move in real income changes the rate by 0.16 births per 1,000, and in marriages by -0.16. The 2025–26 fall was several births per 1,000, so these are small against what they would need to explain. Most of the forecast comes from momentum, not from the drivers.

The core mechanism is rejected. The pre-registered marriage effect has the wrong sign (-0.041, p 0.012), so falsification rule 2 fails. Dropping 2020Q2–2021Q4 leaves it negative but no longer significant (Annex B). The sign is therefore driven largely by the pandemic, when marriages fell in 2020 and births rose in 2021. We report this as the result and do not swap in a specification that passes.

New element. This is the first one-quarter-ahead forecast of the birth rate for Kazakhstan's regions on a harmonised 16-region quarterly panel. Published regional work is annual and descriptive (Aubakirova et al., 2024).

Deviations from pre-registration (memo of 18 October 2026, in `docs/pre_registration.md`):

- Sample starts 2014Q1, estimation from 2016Q1 (pre-registered: 2010Q1). Why: Taldau publishes quarterly births and marriages by region only from January-March 2014. Pre-registered specification: not estimable: no quarterly regional births before 2014.
- Real wage replaced by real per-capita money income (quarterly, deflated by the regional CPI; same lag 4). Why: no quarterly regional wage series was downloaded; per-capita income is the closest quarterly regional income measure obtained. Pre-registered specification: not estimated.
- Housing commissioned replaced by housing per resident (m2, annual, last published year). Why: the housing-commissioned series was not obtained; the housing-stock series measures the same space constraint, more slowly. Pre-registered specification: not estimated.
- Population after January 2024 projected at each region's growth over the previous 12 months. Why: the downloaded population file ends at 1 January 2024. Pre-registered specification: the pre-registration assumed published population for every quarter.
- Chow break dates 2018Q2 / 2020Q2 / 2022Q3 instead of 2015Q3 / 2020Q2 / 2022Q3. Why: 2015Q3 falls before the first estimation quarter (2016Q1); 2018Q2 is the South Kazakhstan / Shymkent split. Pre-registered specification: 2015Q3 not testable inside the sample.
- Driver 'Unemployment rate (yoy change, pp)' dropped. Why: series not available in data/raw. Pre-registered specification: not estimable.
- The memo was written and the model estimated on the same day, and the memo was not time-stamped by the course system before estimation.

4. Validity and limits

Construct. The crude rate measures births per resident, not fertility per woman. Regions with young age structures (Turkestan, Mangystau) look more fertile than their total fertility rate implies, and ageing northern regions look less fertile. Migration of young adults into Astana and Almaty raises the cities' rates and lowers the regions they leave. Projected population after January 2024 biases the latest rates if growth slowed: real growth below the projection makes the rate too low. The January 2023 population revision shifts the denominator by about 2–3% in some regions.

Internal. There is no identification strategy, so the coefficients are labelled conditional correlations, not effects. Omitted variables: family-benefit and mortgage programmes raise income, housing and births together, which biases those coefficients upward. Pandemic shocks moved marriages and births in opposite directions, which biases the marriage coefficient downward. Reverse

causality: an expected birth can bring a marriage forward. The 3-quarter lag reduces this but does not remove it. Measurement error: revisions to the latest data and the projected denominator bias coefficients toward zero. Residual serial correlation with a lagged dependent variable leaves a small bias in the momentum terms.

External. The sample covers 2016–2026: a floating tenge after 2015, COVID-19 in 2020–21, the 2022 border changes, and oil-price swings in a commodity exporter. The Chow tests show the relationship shifted at each of these dates. Results apply to Kazakhstan's high-fertility, oil-dependent regime and do not transfer to low-fertility economies.

Statistical. 8 specifications were run in total, all declared in advance, and every one is reported. An earlier run (2026-10-18, 4 specifications): Same model, with marriages and income entering as annual figures (last published year), run before the quarterly marriage and income files were obtained; marriage coefficient +0.096 (p 0.077, one-sided 0.039); RMSE ratio 0.816; superseded: the quarterly data match the pre-registered specification, so the quarterly run is the main result whatever its sign. With 480 region-quarters the pooled tests have reasonable power. A single region has only 30 quarters, so no region-level effects are claimed. Failing to reject an effect (housing) is not evidence that it is absent.

Robustness check	Marriage coef.	p	RMSE ratio	Winner
Main specification	-0.041	0.012	0.812	model
Alt. sample: drop 2020Q2-2021Q4	-0.036	0.204	0.846	model
Alt. specification: no lagged dependent variable	-0.031	0.355	0.977	model
Alt. measure: births count instead of rate	-0.042	0.014	1.016	benchmark

Overturns the main result: Alt. measure: births count instead of rate. The model's edge over the benchmark depends on the denominator. On births counts the benchmark wins.

5. The product

Three layers with separate entry points. `src/ingest.py` turns raw files into processed data. `src/model.py` writes `model/artifact.json` (coefficients, covariance, diagnostics, benchmark, forecasts). `src/app.py` is a standard-library web server that only reads those two. There is no regression in the request path, and the artifact is re-read on every request. **Run:** `pip install -r requirements.txt` then `python run.py`, which opens http://127.0.0.1:8050. The page shows, for any region or the national total:

- the latest rate, its quarter and the change on a year earlier, above the fold;
- the full history with the source named on the chart;
- each driver's contribution to the forecast;
- a scenario slider whose 90% band comes from the artifact's covariance matrix, with a decision signal;
- a provenance panel and an honesty panel.

Error states: no network shows a labelled cached snapshot; a missing or renamed artifact, missing processed data, or a missing input for a region each give a plain-language message; an edited raw file stops the build. On a fresh clone from GitHub (18 October 2026), acceptance checks 2–10 pass: page in 0.12 s, contributions match the artifact, offline and missing-model messages shown. Checks 4, 5, 8, 9 and 10 are also automated in `tests/`. Check 1 (a clean install on another machine) is left to the Week 6 peer session. A static snapshot is published at i2rchik.github.io/KZ-birth-rates. **Next, with two more weeks:** published 2024–26 population, monthly marriages, quarterly unemployment by region, and an age-standardised fertility measure.

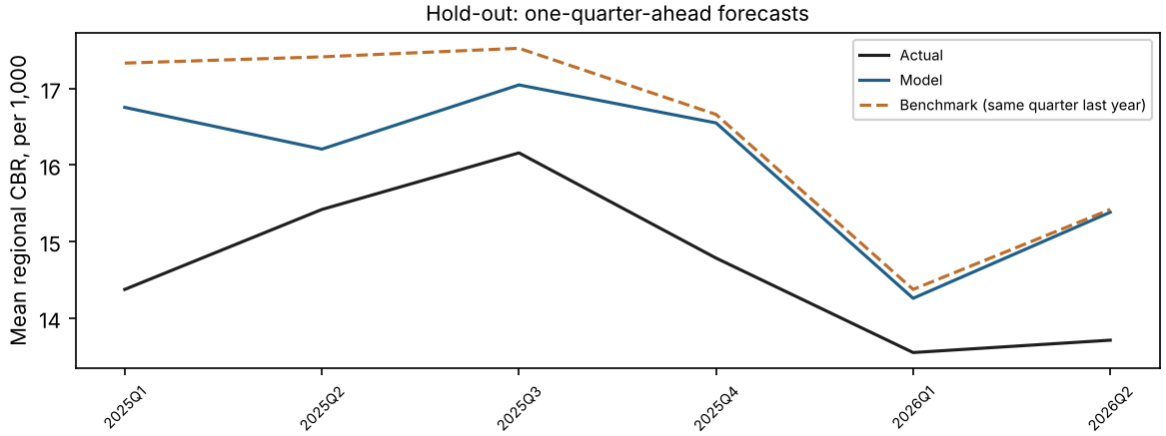
6. Decision rule

User: the maternity and child-health department of a regional akimat, setting next quarter's staffing and bed capacity. **Rule:** if the 90% forecast interval for next quarter lies entirely below (above) the same quarter last year, plan capacity for the forecast percentage fewer (more) births; otherwise plan at last year's level. **Why:** the threshold is the model's own measured uncertainty, a median half-width of ±10.2% across regions. Smaller moves cannot be told apart from noise, and re-staffing on noise is costly. For 2026Q3: 4 regions signal fewer births, 0 signal more, and 12 show no clear signal (Annex C).

Annex A. Full diagnostic and test output

Unit-root tests per region are in `outputs/annex_unit_roots.csv`, cointegration in `outputs/annex_cointegration.csv`, and everything else in `outputs/annex_diagnostics.json`. Monte Carlo p-values use seed 20261018, 2,000 draws. Engle–Granger Fisher p 0.827.

Test	Statistic	p
Serial correlation, lags 1–4 (F, DK)	10.04	<0.001
Breusch–Pagan (LM)	17.38	0.004
Jarque–Bera	28.78	<0.001
Pesaran CD	28.26	<0.001
Chow at 2018Q2 (F, DK)	17.10	<0.001
Chow at 2020Q2 (F, DK)	22.25	<0.001
Chow at 2022Q3 (F, DK)	26.78	<0.001
VIF Own momentum (lag 1)	1.24	
VIF Own momentum (lag 4)	1.31	
VIF Marriage rate (yoy log change), lag 3	1.06	
VIF Real per-capita income (yoy log change), lag 4	1.02	
VIF Housing per resident, last published year (yoy log change), lag 1	1.03	



Hold-out RMSE by region

Region	RMSE model	RMSE benchmark
Akmola	1.213	1.472
Aktobe	1.963	2.316
Almaty city	1.488	1.609
Almaty oblast (incl. Zhetisu)	1.916	2.291
Astana	1.560	2.158
Atyrau	1.866	2.529
East Kazakhstan (incl. Abai)	1.239	1.616
Karaganda (incl. Ulytau)	1.161	1.596
Kostanay	0.772	1.115
Kyzylorda	2.402	2.711
Mangystau	2.287	3.176
North Kazakhstan	0.566	0.875
Pavlodar	0.706	1.010
Turkestan + Shymkent	2.614	2.818
West Kazakhstan	1.588	1.864
Zhambyl	1.715	2.095

Annex B. Robustness (full)

Check	Obs.	Marriage coef.	p	RMSE ratio	Winner
Main specification	480	-0.041	0.012	0.812	model
Alt. sample: drop 2020Q2-2021Q4	368	-0.036	0.204	0.846	model
Alt. specification: no lagged dependent variable	544	-0.031	0.355	0.977	model
Alt. measure: births count instead of rate	480	-0.042	0.014	1.016	benchmark

Annex C. Next-quarter forecasts and decision signals

Region	Same quarter last year	Forecast	90% interval	Signal
Akmola	11.91	11.30	10.26–12.45	no clear signal: plan at last year's level
Aktobe	17.20	15.99	14.50–17.62	no clear signal: plan at last year's level
Almaty city	14.85	13.55	12.29–14.93	no clear signal: plan at last year's level
Almaty oblast (incl. Zhetisu)	18.50	17.48	15.86–19.28	no clear signal: plan at last year's level
Astana	17.86	17.25	15.66–18.99	no clear signal: plan at last year's level
Atyrau	19.83	17.12	15.49–18.93	plan for 13.6% fewer births
East Kazakhstan (incl. Abai)	12.85	12.22	11.10–13.46	no clear signal: plan at last year's level
Karaganda (incl. Ulytau)	13.62	12.56	11.39–13.84	no clear signal: plan at last year's level
Kostanay	10.32	9.71	8.81–10.69	no clear signal: plan at last year's level
Kyzylorda	20.86	19.04	17.27–20.99	no clear signal: plan at last year's level
Mangystau	21.75	19.97	18.12–22.02	no clear signal: plan at last year's level
North Kazakhstan	9.37	8.49	7.71–9.35	plan for 9.4% fewer births
Pavlodar	11.90	10.95	9.94–12.06	no clear signal: plan at last year's level
Turkestan + Shymkent	22.95	20.67	18.72–22.82	plan for 9.9% fewer births
West Kazakhstan	15.42	13.87	12.58–15.29	plan for 10.1% fewer births
Zhambyl	19.33	17.68	16.04–19.48	no clear signal: plan at last year's level

Annex D. References (APA 7)

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Each link must be opened and checked by a team member before submission (brief, Section 15).

Annex E. AI-use declaration

See `AI_USE.md`. Claude (Anthropic) wrote the first version of the code, the documents and this note's wording, and searched for references. All numbers come from the code's outputs, not from the AI's text. The team must verify the items listed in `AI_USE.md` and sign it.

Annex F. Contribution statement and replication receipt

See `CONTRIBUTIONS.md` (signed by every member) and `docs/replication_receipt.md` (signed by the reviewing team in Week 6).